FARM MANAGEMENT SYSTEM

Salesforce-Based Project

TEAM MEMBERS

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1. Introduction

1.1 Project Overview

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards.

The system helps farmers and farm managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose

The main purposes of this project are:

- To digitize agricultural data management.
- To improve data accuracy and accessibility.
- To automate record handling.
- To enable real-time monitoring.
- To support data-driven farming decisions.
- To enhance productivity and transparency.

2. Ideation Phase

2.1 Problem Statement

ID	Customer	Trying to	Problem	Because	Feeling
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are maintained manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Administrator	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

2.2 Empathy Map Canvas

Target Users: Farmers, Farm Managers, Buyers, System Administrators

Aspect	Description
Think & Feel	Need accurate and timely farm data.
Hear	Feedback from agricultural officers.
See	Registers, paper files, and spreadsheets.
Say & Do	Enter records manually.

Aspect	Description
Pain	Data loss, errors, and delays.
Gain	Automation and fast reporting.

2.3 Brainstorming

The following ideas were considered:

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm application
- Salesforce-based Farm Management System

Selected Idea: Salesforce-based Farm Management System was selected as the final solution.

3. Requirement Analysis Phase

3.1 Customer Journey Map

Login > Access Farm Management App > Enter Farm and Crop Data > Register Farmers and Buyers > Generate Reports > Analyze Results > Decision Making

3.2 Functional Requirements

ID	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer and Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation
FR-6	Dashboard Visualization
FR-7	User Role Management

3.3 Non-Functional Requirements

Requirement	Description
Usability	Easy interface
Security	Data protection
Reliability	No data loss
Performance	Fast access
Availability	24/7 access
Scalability	Supports more farms and users

3.4 Data Flow Diagram

User > Salesforce UI > Custom Objects > Validation Rules > Relationships > Reports > Dashboards

The main custom objects are: Farm, Crop, Farmer, Buyer.

4. Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

5. Project Design Phase

5.1 Problem-Solution Fit

The Farm Management System replaces manual farming records with automated data management. Cloud storage helps prevent data loss, dashboards improve analysis, and real-time reports reduce delays.

5.2 Proposed Solution

The proposed solution is a Salesforce-based Farm Management System with centralized data storage and automation.

Innovation

- Automated data-entry validation
- Cloud storage
- Real-time reporting
- Interactive dashboards

Impact

- Improved data accuracy
- Faster reporting
- Better decision-making

Business Model: Digital farm management and agricultural data services platform.

Scalability: The system supports multiple farms, users, and regions through cloud-based infrastructure.

6. Solution Architecture

The system contains four main layers:

Layer	Component
1. Presentation Layer	Salesforce Lightning UI
2. Application Layer	Validation Rules and Relationships
3. Data Layer	Custom Objects
4. Analytics Layer	Reports and Dashboards

7. Project Planning & Scheduling

The project activities include:

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

7.1 Team Contribution

Team Member	Responsibility
Kalaiselvan S	Team Lead, Project Coordination & Integration
Praveen N	Salesforce Development
Harinath M	Data Modeling & Configuration
Dharshini V	Security, Profiles & Roles
Vvishnu	Reports, Dashboards & Testing

8. Product Backlog & Sprint Plan

Sprint	Functional Requirement	User Story	Points	Priority
Sprint-1	Developer Setup	Create and configure Salesforce Developer Org	3	High
Sprint-2a	Data Modeling	Create Farm, Crop, Farmer and Buyer objects	5	High
Sprint-2	Data Modeling	Create tabs and Lightning App	5	High
Sprint-3	Automation	Validate farm and crop details	3	High
Sprint-3	Automation	Automate crop updates and buyer notifications	3	High
Sprint-4	Security	Implement role-based access control	5	High

Sprint	Functional Requirement	User Story	Points	Priority
Sprint-5	Reports	Create crop yield, sales and buyer reports	4	High
Sprint-6	Dashboards	Create productivity and sales dashboards	4	Medium

9. Project Development Phase

9.1 Salesforce Developer Org

A Salesforce Developer Org was created to develop and configure the Farm Management System. The Developer Org provides the required Salesforce features for developing the application.

10. Custom Objects

The application uses four custom objects: Farm, Crop, Farmer and Buyer. Salesforce objects work as database tables containing fields and records.

11. Farm Object

The Farm object contains: Farm Name, Farm Type, Soil Quality, Irrigation Type, Location, Size, Farmer Relationship.

Field	Values
Farm Type	Crop, Livestock, Mixed
Soil Quality	Organic, Cover Crops, Composting
Irrigation Type	Sprinkler irrigation, Drip irrigation, Localized irrigation

A Master-Detail relationship is created with the Farmer object.

12. Crop Object

The Crop object contains: Crop Name, Planting Date, Harvest Date, Farm Relationship, Actual Yield, Total Acres Harvested, Expected Yield per Acre.

A Lookup relationship connects Crop with Farm. The Expected Yield per Acre is calculated using: **Actual Yield ÷ Total Acres Harvested**

13. Farmer Object

The Farmer object contains: Contact Number, State, District, Village. A field dependency is created between State and District, so the appropriate district values are displayed based on the selected state.

14. Buyer Object

The Buyer object contains: Contact, Email, Address, Buyer Type. Buyer Type values: Wholesale, Retail.

15. Custom Tabs

Custom tabs are created for Farm, Crop, Farmer and Buyer. These tabs allow users to access the respective object records easily.

16. Lightning App

A Lightning App named **Farm Management System** is created. The application includes Farm, Crop, Farmer, Buyer, Reports and Dashboards. The application is configured for the System Administrator profile.

17. Profile Management

A profile defines the permissions and settings that determine what a Salesforce user can access and perform. A custom Farm Manager profile is created for the project.

18. Role Management

A Farm Manager role is created to establish the reporting hierarchy and manage access to relevant data.

19. User Management

A Farm Manager user is created with:

Setting	Value
First Name	Farm
Last Name	Manager
Role	Farm Manager
User License	Standard Platform User
Profile	Farm Manager

20. Sharing Rules

Sharing Rules are used to share records with users based on specific conditions. A criteria-based sharing rule is configured using the Can Join Immediately condition from the Crop object.

Setting	Value
Criteria	Crop = True
Shared With	Farm Manager
Access Level	Read/Write

21. User Adoption

A Farmer record can be created through the Farm Management application:

1. Open App Launcher.
2. Search Farm Management.
3. Open the application.

4. Select Farmer.
5. Click New.
6. Enter the required details.
7. Save the record.

22. Reports

Reports display records based on defined criteria. They can contain rows, columns, filters, groups, and graphical charts. A report named **Farmers with Farms** is created and saved in a public folder. Another report can be created using Crops and Buyers objects.

23. Dashboard

A dashboard named **Farmers with Farms** is created. Dashboard configuration:

1. Open Dashboard.
2. Click New Dashboard.
3. Enter Farmers with Farms.
4. Click Create.
5. Add a Component.
6. Select Farmers with Farms report.
7. Select Gauge chart.
8. Click Add.
9. Save the dashboard.

24. Functional & Performance Testing

The following features were tested:

- Farm, Crop, Farmer and Buyer record creation
- Input validation
- Relationship checks
- Field dependency checks
- Profile and role access
- Sharing rule verification
- Report and dashboard generation

Performance testing was also conducted to check system response and stability during multiple user operations.

25. Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy

- Better monitoring
- Real-time analytics

26. Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

27. Conclusion

The Farm Management System successfully digitizes agricultural data management using Salesforce. The system helps improve efficiency, reduce errors, and provide useful agricultural information through automation, reports, and dashboards. It provides a centralized platform for managing farms, crops, farmers, and buyers.

28. Future Scope

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

29. Appendix

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

30. Project Demonstration

During the project demonstration, the following modules can be shown:

- Salesforce Login
- Farm Management System App
- Farm, Crop, Farmer and Buyer Objects
- Fields & Relationships
- Profiles, Roles and Users
- Sharing Rules
- Reports and Dashboards
- Testing Results

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THANK YOU